

## EXERCISE-1

### Creating and Managing Tables

#### OBJECTIVE

After the completion of this exercise, students should be able to do the following:

- Create tables
- Describing the data types that can be used when specifying column definition
- Alter table definitions
- Drop, rename, and truncate tables

#### NAMING RULES

Table names and column names:

- Must begin with a letter
- Must be 1-30 characters long
- Must contain only A-Z, a-z, 0-9, \_, \$, and #
- Must not duplicate the name of another object owned by the same user
- Must not be an oracle server reserve words
- 2 different tables should not have same name.
- Should specify a unique column name.
- Should specify proper data type along with width
- Can include "not null" condition when needed. By default it is 'null'.

#### The CREATE TABLE Statement

**Table:** Basic unit of storage; composed of rows and columns

**Syntax: 1** Create table table\_name (column\_name1 data\_type (size)  
column\_name2 data\_type (size)...);

**Syntax: 2** Create table table\_name (column\_name1 data\_type (size) constraints,  
column\_name2 data\_type constraints ...);

#### Example:

Create table employees ( employee\_id number(6), first\_name varchar2(20), ..job\_id varchar2(10),  
CONSTRAINT emp\_emp\_id\_pk PRIMARY KEY (employee\_id));

#### Tables Used in this course

#### Creating a table by using a Sub query

#### **SYNTAX**

// CREATE TABLE table\_name(column\_name type(size)...);

Create table table\_name as select column\_name1,column\_name2,.....column\_namen from  
table\_name where predicate;

#### **AS Subquery**

Subquery is the select statement that defines the set of rows to be inserted into the new table.

### Example

Create table dept80 as select employee\_id, last\_name, salary\*12 Annsal, hire\_date  
from employees where dept\_id=80;

### The ALTER TABLE Statement

The ALTER statement is used to

- Add a new column
- Modify an existing column
- Define a default value to the new column
- Drop a column
- To include or drop integrity constraint.

### SYNTAX

ALTER TABLE table\_name ADD /MODIFY(Column\_name type(size));

ALTER TABLE table\_name DROP COLUMN (Column\_name);

ALTER TABLE ADD CONSTRAINT Constraint\_name PRIMARY KEY (Column\_Name);

### Example:

Alter table dept80 add (job\_id varchar2(9));

Alter table dept80 modify (last\_name varchar2(30));

Alter table dept80 drop column job\_id;

**NOTE:** Once the column is dropped it cannot be recovered.

### DROPPING A TABLE

- All data and structure in the table is deleted.
- Any pending transactions are committed.
- All indexes are dropped.
- Cannot roll back the drop table statement.

### Syntax:

Drop table *tablename*;

### Example:

Drop table dept80;

### RENAMING A TABLE

To rename a table or view.

### Syntax

RENAME old\_name to new\_name

**Example:**

Rename dept to detail\_dept;

**TRUNCATING A TABLE**

Removes all rows from the table.

Releases the storage space used by that table.

**Syntax**

TRUNCATE TABLE *table\_name*;

**Example:**

TRUNCATE TABLE copy\_emp;

**Find the Solution for the following:**

Create the following tables with the given structure.

**EMPLOYEES TABLE**

| NAME           | NULL?    | TYPE        |
|----------------|----------|-------------|
| Employee_id    | Not null | Number(6)   |
| First_Name     |          | Varchar(20) |
| Last_Name      | Not null | Varchar(25) |
| Email          | Not null | Varchar(25) |
| Phone_Number   |          | Varchar(20) |
| Hire_date      | Not null | Date        |
| Job_id         | Not null | Varchar(10) |
| Salary         |          | Number(8,2) |
| Commission_pct |          | Number(2,2) |
| Manager_id     |          | Number(6)   |
| Department_id  |          | Number(4)   |

**DEPARTMENT TABLE**

| NAME        | NULL?    | TYPE        |
|-------------|----------|-------------|
| Dept_id     | Not null | Number(6)   |
| Dept_name   | Not null | Varchar(20) |
| Manager_id  |          | Number(6)   |
| Location_id |          | Number(4)   |

**JOB\_GRADE TABLE**

| NAME        | NULL? | TYPE       |
|-------------|-------|------------|
| Grade_level |       | Varchar(2) |
| Lowest_sal  |       | Number     |

|             |        |
|-------------|--------|
| Highest sal | Number |
|-------------|--------|

## LOCATION TABLE

| NAME           | NULL?    | TYPE        |
|----------------|----------|-------------|
| Location_id    | Not null | Number(4)   |
| St_addr        |          | Varchar(40) |
| Postal_code    |          | Varchar(12) |
| City           | Not null | Varchar(30) |
| State_province |          | Varchar(25) |
| Country_id     |          | Char(2)     |

1. Create the DEPT table based on the DEPARTMENT following the table instance chart below. Confirm that the table is created.

| Column name  | ID     | NAME     |
|--------------|--------|----------|
| Key Type     |        |          |
| Nulls/Unique |        |          |
| FK table     |        |          |
| FK column    |        |          |
| Data Type    | Number | Varchar2 |
| Length       | 7      | 25       |

```
CREATE TABLE DEPTC
  ID NUMBER(7) CONSTRAINT dept_id_pk primary key,
  NAME VARCHAR(25) NOT NULL
TABLE CREATED
```

2. Create the EMP table based on the following instance chart. Confirm that the table is created.

| Column name  | ID     | LAST_NAME | FIRST_NAME | DEPT_ID |
|--------------|--------|-----------|------------|---------|
| Key Type     |        |           |            |         |
| Nulls/Unique |        |           |            |         |
| FK table     |        |           |            |         |
| FK column    |        |           |            |         |
| Data Type    | Number | Varchar2  | Varchar2   | Number  |
| Length       | 7      | 25        | 25         | 7       |

```
CREATE TABLE EMP
  ID NUMBER(7) CONSTRAINT emp_id_pk primary key,
  FIRST_NAME VARCHAR(25) NOT NULL,
  DEPT_ID NUMBER(7),
  CONSTRAINT emp_dept_fk FOREIGN KEY (DEPT_ID) REFERENCE (DEPT)
```

3. Modify the EMP table to allow for longer employee last names. Confirm the modification. (Hint: Increase the size to 50)

```
ALTER TABLE EMP
  MODIFY LAST_NAME VARCHAR2(50);
```

4. Create the EMPLOYEES2 table based on the structure of EMPLOYEES table. Include Only the Employee\_id, First\_name, Last\_name, Salary and Dept\_id columns. Name the columns Id, First\_name, Last\_name, salary and Dept\_id respectively.

```
CREATE TABLE EMPLOYEES2 AS  
SELECT  
    Employee_id AS ID,  
    First_name ,  
    Last_name ,  
    salary ,  
    Department_id AS DEPT_id  
FROM EMPLOYEES;
```

5. Drop the EMP table.

```
DROP table EMPLOYEES;  
Table dropped.
```

6. Rename the EMPLOYEES2 table as EMP.

```
RENAME EMPLOYEES2 to EMP
```

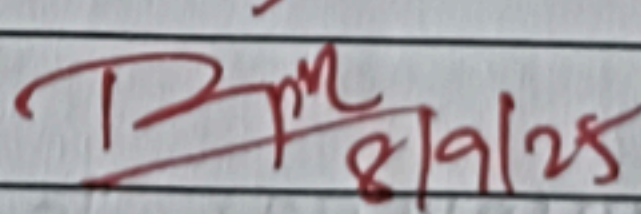
7. Add a comment on DEPT and EMP tables. Confirm the modification by describing the table.

COMMENT ON TABLE EMP IS 'Employee details table linked to DEPT table via department ID's'

8. Drop the First\_name column from the EMP table and confirm it.

ALTER TABLE EMP

DROP ~~COLUMN~~ FIRST NAME;

| Evaluation Procedure | Marks awarded                                                                                   |
|----------------------|-------------------------------------------------------------------------------------------------|
| Query(5)             | 5                                                                                               |
| Execution (5)        | 5                                                                                               |
| Viva(5)              | 5                                                                                               |
| Total (15)           | 15                                                                                              |
| Faculty Signature    | <br>8/9/25 |